

Box 1: Emergency planning zone (EPZ)

EPZ is the designated area around the nuclear reactor where emergency response measures and procedures have been planned to protect the public and property in case of an accident or emergency. The size of the EPZ depends on the type of reactor, potential hazards, and the applicable regulations. For large nuclear reactors, EPZs typically range from 16 to 80km. For SMRs, EPZs are significantly smaller, sometimes limited to the plant site itself, making them suitable for deployment near industrial sites and urban areas.

With the market projected to reach \$300 billion by 2040, SMRs are set to redefine nuclear energy and play a pivotal role in shaping a sustainable, resilient power landscape. Several companies and startups across various countries are actively developing SMR technologies.

How they work

SMRs use the same fundamental nuclear fission process as conventional large-scale reactors to generate heat, which powers a turbine to produce electricity. Key components include a reactor core, coolant system, steam generator, turbine, electric generator, condenser, and passive safety mechanisms.

At the heart of an SMR lies the reactor core, where the controlled fission reaction takes place, releasing a large quantity of heat. This heat is transferred through a coolant system to the steam generator, where it converts water into high-pressure steam that drives a turbine connected to a generator. The used steam enters a condenser, where it is cooled into water and recirculated to continue the energy production cycle. The reactor core uses control rods made of neutron-absorbing materials, such as boron, to regulate the fission process and prevent overheating. The coolant systems may use light water, gas, liquid metal, or molten salt. Unlike conventional reactors, which use electrically powered active cooling systems, SMRs rely on natural convection and gravity-driven mechanisms to maintain reactor cooling, even during power failures

or emergencies, thereby reducing the risk of overheating.

SMRs design

SMRs are designed for flexibility, scalability and efficiency. Emphasis is placed on compactness and factory assembly to enable standardised production, enhanced reliability, reduced maintenance needs, and a lower risk of human error. Their designs typically feature streamlined architecture, integrated systems, fewer components, and advanced digital monitoring technologies. Prefabricated as modular units, SMRs are easily transported via road, rail, or ship, enabling quick deployment in remote locations where building large-scale facilities is impractical.

Refuelling is required every three to seven years, compared to one to two years for conventional reactors. Some advanced designs can operate continuously for up to 30 years without refuelling. Improved fuel recycling and greater efficiency help to minimise radiation levels and reduce radioactive waste.

These designs incorporate advanced manufacturing methods, such as additive manufacturing, 3D printing, powder metallurgy, and hot isostatic pressing (HIP), to enhance material durability and enable reactors to withstand extreme thermal and nuclear conditions. Critical materials are used to improve performance and longevity. For example, zirconium alloy offers superior corrosion resistance while minimising neutron absorption, and nickel-based superalloys improve heat resistance while reducing

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